• **STEERING—STEERING WHEEL "NIBBLE"—SPEEDS ABOVE 100 KM/H (60 MPH)** • **NOISE—LOW FREQUENCY "HUM" AND/OR "BOOM"—SPEEDS ABOVE 100 KM/H (60 MPH)** • **VIBRATION—FELT IN SEAT, STEERING WHEEL AND/OR FLOORPAN—SPEEDS ABOVE 100 KM/H (60 MPH)**	**Article No. 01-15-1**

LINCOLN: 2000-2002 LS

This article is being republished in its entirety to update the model year coverage.

ISSUE

Some vehicles may exhibit one of the following types of vehicle or steering wheel vibrations:

- Steering wheel "nibble" - A side-to-side rotational oscillation of the steering wheel that is typically most noticeable at vehicle speeds above 100 km/h (60 mph)
- Steering wheel vibration and/or vehicle vibration - A "buzzing" or high frequency sensation in the steering wheel and/or in the seat and floorpan, typically most noticeable at speeds above 100 km/h (60 mph)
- Audible low frequency "hum"/"boom" - A low frequency hum or boom that is most noticeable at speeds above 100 km/h (60 mph)

ACTION

Refer to the following detailed evaluation, diagnostic, and repair procedures.

VEHICLE EVALUATION PROCEDURE

1. Inspect vehicle suspension, driveline, wheels/tires and steering system for any damaged, loose, or missing components.
2. Verify that the tire pressures for all four tires are inflated to 30 psi (210 kPa) when cold.

NOTE

DUE TO THE TIRE CONSTRUCTION OF THE HIGH-SPEED-RATED, HANDLING-ORIENTED TIRES USED ON THE LS, IT MAY BE NECESSARY TO DRIVE THE VEHICLE FOR 8-16 KILOMETERS (5-10 MILES) AT HIGHWAY SPEEDS PRIOR TO EVALUATING THE VEHICLE FOR ANY TYPE OF VIBRATION. THIS WILL WORK OUT ANY FLAT SPOTTING THAT MAY HAVE OCCURRED IN THE TIRES, ESPECIALLY IF THE VEHICLE HAS BEEN PARKED FOR AN EXTENDED PERIOD OF TIME.

3. Drive the vehicle to evaluate and confirm the condition. Note the speeds where the condition exists and if the vehicle exhibits vibration and/or boom (audible hum or drone) or if the condition is steady or phases in and out when driving at a steady speed. If the condition was present at the beginning of the drive and went away after driving several kilometers/miles, the condition was due to flat spotting of the tires and no further service actions are required. Tire flat spotting is a normal characteristic of the high-speed-rated, performance-oriented tires installed on the LS. When flat spotting occurs, it will usually work itself out after the first few kilometers/miles of driving the vehicle. If the condition is still present, refer to the flowchart (Figure 1) for the proper service actions.

PROCEDURE 1: INTERMEDIATE STEERING SHAFT (I-SHAFT) NEUTRALIZATION

1. Loosen the I-shaft slider locking bolt located under the vehicle between the dash boot bearing and the lower steering shaft U-joint. It may be necessary to rotate the steering shaft to gain access to the bolt. Do not loosen the pinch bolts at either ends of the I-shaft, loosen the slider bolt only.
2. Remove the trim panel from under the driver's side instrument panel.
3. Remove the intermediate shaft boot bearing nuts that secure the boot bearing to the dash panel.

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4. Slide the boot bearing up the I-shaft and spin it to verify the bearing/seal assembly rotates freely and smoothly.
5. Reseat the boot bearing assembly to the dash panel and torque the nuts to 9 N•m (80 lb-in).
6. Reinstall the trim panel under the instrument panel.
7. Tighten the I-shaft slider bolt to 35 N•m (26 lb-ft).

PROCEDURE 2: STEERING GEAR YOKE SPRING REPLACEMENT

1. Make sure vehicle is steering straight ahead. Failure to have tires in the straight ahead position will result in improper force on the yoke spring when the wheels are steered straight.
2. Loosen the locknut that secures the yoke cap.
3. Remove the yoke cap. Use care to ensure that the yoke spring and yoke body do not fall out of the rack assembly when the yoke cap is removed. Do not remove the yoke body if it does not fall out of the rack, it is important that the alignment of the yoke body and steering rack is maintained when the yoke body is placed back into the rack.
4. Remove the 25# yoke spring (green color).
5. Install new 35# Yoke Spring (EOBZ-3F516-A white color).
6. Replace the yoke cap and tighten to 7 N•m (60 lb-in). The correct torque is critical and an accurate torque wrench must be used.
7. After torquing the yoke cap to the correct specification, loosen the cap by turning the yoke cap 45 degrees (1/8 turn). This sets the proper pre-load on the yoke spring.
8. While holding the yoke cap, tighten the locknut.

To obtain OEM quality tires, refer to the following tire manufacturer phone numbers:

- Bridgestone/Firestone - U.S. and Canada - 1-800-356-4644
- Continental - U.S. - 1-800-847-3349
- Continental - Canada - 1-800-461-1776

PART NUMBER	PART NAME
EOBZ-3F516-A	Yoke Spring

OTHER APPLICABLE ARTICLES: NONE
SUPERSEDES: 00-3-4
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 303000, 304000, 306000, 703000, 703200, 703300, 703400

LS STEERING WHEEL VIBRATION/NIBBLE DIAGNOSTIC PROCEDURE

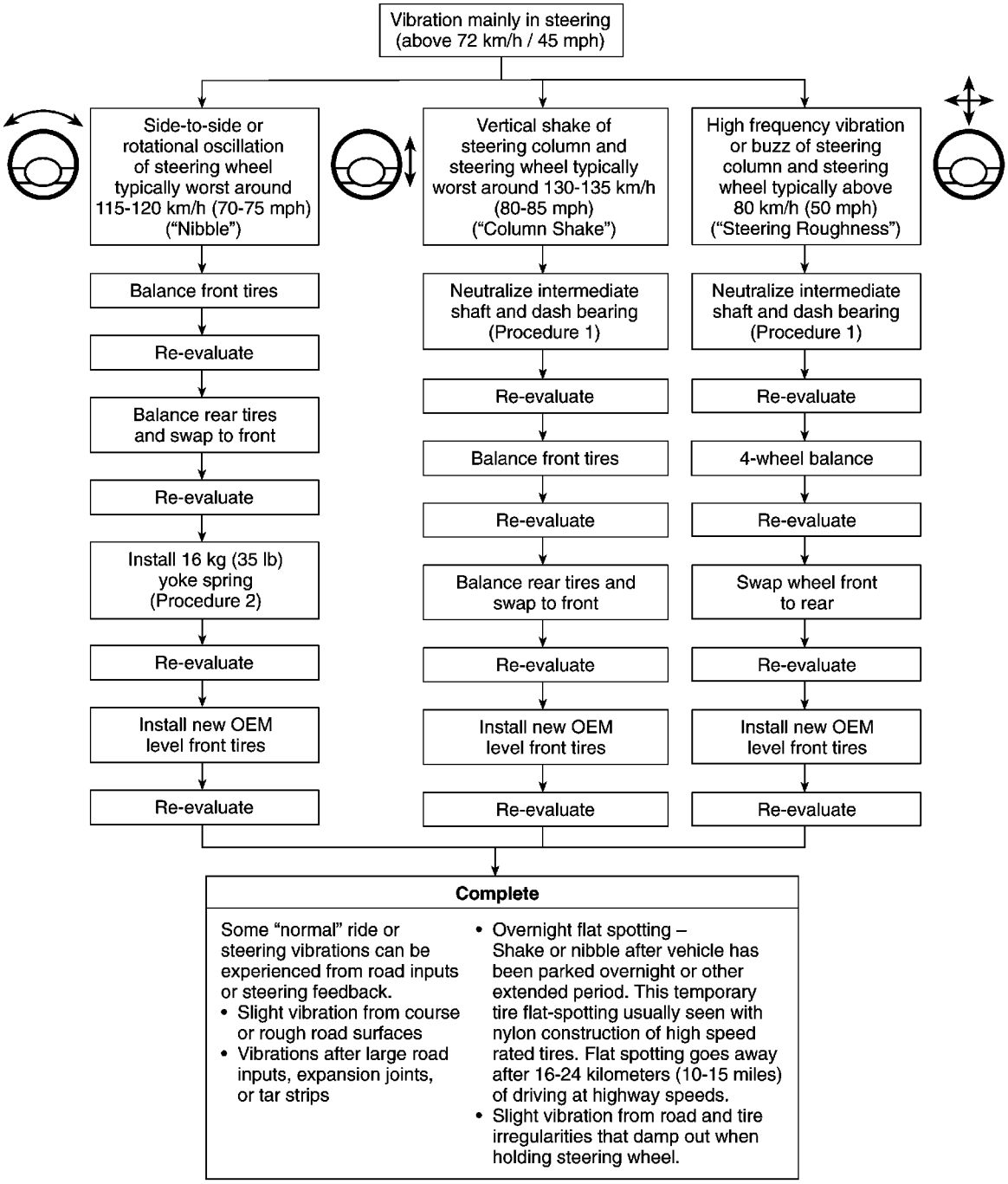

Figure 1 - Article 01-15-1